P.T. INTI KARYA PERSADA TEHNIK ENGINEERING & CONSTRUCTION KELLOGG OVERSEAS CORPORATION	DOC. No. DS-H04-K-001 Job No. : 95038 / 6413	SHEET No. : 1 OF 14

DATA SHEET :

PROPANE COMPRESSOR
H4K-1

A		12/6/98
REV NO.	APPROVED BY	DATE
FINAL APPROVAL BY PKP		

CLIENT : PERUSAHAAN PERTAMBANGAN MINYAK DAN GAS BUMI NEGARA (PERTAMINA)/ PROYEK KILANG PENGOLAHAN (PKP)

JOB TITLE : BONTANG LNG EXPANSION PROJECT TRAIN H

PROJECT NAME : PGH PROJECT

LOCATION : BONTANG, EAST KALIMANTAN, INDONESIA

FOR RECORD

REV	DATE	PAGE	DESCRIPTION	PREP'D	CHK'D	APP'D
0	19 Dec.1996	ALL	Issued for Approval and Inquiry	ADN SDP	AS	BS JGB
1	29 Jan. 1997	ALL	For PKP review during Bid Clarification Meeting	ADN SDP	AS SWK	
2	6 Feb. 1997	ALL	Issued for Approval based on PKP comments at Bid Clarification Meeting	ADN IRA	AS SWK	BS JGB
3	24 Mar. 1997	ALL	Issued for PKP Approval revised based on PKP comments HX-PJIP-0049	ADN	AS SWK	BS JGB
4	16 Apr. 1997	2	Approved per HX-PJIP-0112/0165 and issued for Purchase Order	ADN	AS SWK	BS JGB
5	7 Jan. 1998	2,3,4,5,6 &13,14	Revised reflected 3rd & 4th impeller changing and reissued for stamp & sign	ADN HNY	AS TLM	BS JGB
6	25-May-98	2,3,8	Revised per data from DR and reissued for stamp & sign			

IKPT KELLOGG

CENTRIFUGAL COMPRESSOR DATA SHEET SI (METRIC) UNITS

JOB NO. 95038 / 6413 ITEM NO. H4K-1
P.O. NO. PGH-KPO-K-003 DATE 25 May 1998
REQUISITION NO. PGH-MR-K-003
DOC. NO. DS-H04-K-001

APPLICABLE TO : PROPOSAL ● PURCHASE AS BUILT DATE 25 May 1998 REVISION 6
FOR BONTANG LNG EXPANSION PROJECT, TRAIN-H UNIT PLANT 4 - REFRIGERATION
SITE BONTANG, EAST KALIMANTAN, INDONESIA SERIAL NO.
SERVICE PROPANE COMPRESSOR NO. REQUIRED ONE
MANUFACTURER DRESSER - RAND MODEL 939 B6-4 DRIVER STEAM TURBINE (H4KT-1)
NOTE : INFORMATION TO BE COMPLETED : BY PURCHASER BY MANUFACTURER.

OPERATING CONDITIONS

(ALL DATA ON PER UNIT BASIS)

	DESIGN & GUARANTEE NORMAL - CASE 1			ALT. OPERATION CASE 2		
	Low press	Med. press	Hi. press	Low press	Med. press	Hi. press
	INLET	SS # 1	SS # 2	INLET	SS # 1	SS # 2
	Propane	Propane	Propane	Propane	Propane	Propane
◉ GAS HANDLED (Also See Page 4) nm3/h (1.013 bar & 0°C Dry)						
◉ Weight Flow, kg/h (Dry)	589,731	1,130,473	1,502,805	616,538	1,149,913	1,512,297
INLET CONDITIONS						
◉ Pressure, (Kg/cm2) A NOTE (3)	1.19	3.75	7.35	1.26	3.83	7.36
◉ Temperature, °C NOTE (3)	-37.84	- 6.72	16.28	-36.68	- 6.0	16.7
◉ Mix Temperature °C		+ 1.9	22.7		2.4	22.9
◉ Molecular Weight, M	44.097	44.097	44.097	44.097	44.097	44.097
◉ Cp / Cv (K1) (SS#1 &SS#2-Inlet/Aftermix) Note 2	1.15	1.148	1.132	1.15	1.148	1.132
☒ Compressibility (Z1) (SS#1 &SS#2-Inlet/Aftermix) Note 2	.966	.918 / .927	.868 / .884	.965	.916 / .926	.867 / .882
☒ Inlet Volume, m3/h (Dry)	216,835	148,093	102,488	214710	147,376	100,780
DISCHARGE CONDITIONS				SECT.1	SECT.2	SECT.3
☒ Pressure, (Kg/cm2) A NOTE (3)	3.75	7.35	14.50	3.83	7.36	14.50
☒ Temperature, °C (to be verified by vendor)	9.3	24.8	59.5	9.1	24.8	59.4
☒ Cp / Cv (K2) AVG	1.150	1.145	1.142	1.150	1.145	1.142
☒ Compressibility (Z2) AVG	.951	.912	.844	.948	.896	.836
☒ KW Required (All Losses Included) ± 4 %		39,489			39,384	
☒ Speed, rpm		3,058			3,026	
☒ Estimated Surge, m3/h (at Speed Above)	SEE PERFORMANCE CURVES			SEE PERFORMANCE CURVES		
☒ Polytropic Head, M-kg/kg F.F / IMP.	5416/5583	3379/2735	3465/3946	5251/5408	3241/2642	3416/3864
☒ Polytropic Efficiency, % F.F / IMP.	.824/.850	1.051/.859	.720/.819	.822/.847	1.052/.857	.725/.820
◉ Guarantee Point See note 2, page 13 of 15	YES	YES	YES			
☒ Performance Curve No	see pages	4, 5, 6, 7 of 12				

PROCESS CONTROL By HCJV

Method : ○ Suction Throttling From ____ (bar) To ____ (bar) ○ Variable Inlet Guide Vanes (3.4.2.4) ◉ Speed Variation From 80 % To 105 % ○ Discharge Blowoff To ____ ◉ Cooled Bypass From Discharge To Suction

Signal : ◉ Source (3.4.2.1) MANUAL CONTROLLER (Electronic Signal/Hydraulically Actuated) FOR SPEED CONTROL

Type : ○ Electronic ○ Pneumatic ◉ Other ANTI-SURGE CONTROLLER (Electronic Signal / Pneumatically Actuated) FOR COOLED BYPASS

Range 4 - 20 mA 0.2 - 1.0 (kg/cm2G)

Antisurge Bypass : ○ Manual ◉ Automatic ○ None

REMARKS

(1) AT SIDESSTREAM INLET TO COMPRESSOR (APPLIES TO MED. & HI. PRESS. COLUMNS)
(2) VENDOR SHALL CONFIRM GAS PROPERTIES AT EACH INLET, EACH DISCHARGE AND MIX TEMPERATURES AT SS#1 AND SS#2 INLETS
(3) CASE 1 : MINIMUM BUTANE RECYCLE. CASE 2 : MAXIMUM BUTANE RECYCLE. CASE 3 : NO LPG PRODUCTION
(4) DELETED

IKPT — KELLOGG

CENTRIFUGAL COMPRESSOR DATA SHEET SI (METRIC) UNITS

JOB NO. 95038 / 6413 ITEM NO. H4K-1
P.O. NO. PGH-KPO-K-003 DATE 25 May 1998
REQUISITION NO. PGH-MR-K-003
DOC. NO. DS-H04-K-001

APPLICABLE TO : PROPOSAL • PURCHASE AS BUILT DATE 25 May 1998 REVISION 6
FOR BONTANG LNG EXPANSION PROJECT, TRAIN-H UNIT PLANT 4 - REFRIGERATION
SITE BONTANG, EAST KALIMANTAN, INDONESIA SERIAL NO.
SERVICE PROPANE COMPRESSOR NO. REQUIRED ONE
MANUFACTURER DRESSER - RAND MODEL 939 B6-4 DRIVER STEAM TURBINE (H4KT-1)
NOTE : INFORMATION TO BE COMPLETED : BY PURCHASER BY MANUFACTURER.

OPERATING CONDITIONS

(ALL DATA ON PER UNIT BASIS)

	ALT. OPERATION CASE-3			ALT. RECYCLE CASE (3) (BY) VENDOR		
	Low press	Med. press	Hi. press	Low press	Med. press	Hi. press
	INLET	SS # 1	SS # 2	INLET	SS # 1	SS # 2
◉ GAS HANDLED (Also See Page 4) nm3/h (1.013 bar & 0°C Dry)	Propane	Propane	Propane			
◉ Weight Flow, kg/h (Dry)	580,671	1,124,834	1,497,513	641,255	351,241	167,897
INLET CONDITIONS						
◉ Pressure, (Kg/cm2) A	1.17	3.71	7.33	1.90	4.75	7.58
◉ Temperature, °C	-38.27	- 7.0	16.0	41.9	41.9	41.9
◉ Mix Temperature °C		1.5	22.6		67.6	80.6
◉ Molecular Weight, M	44.097	44.097	44.097	44.097	44.097	44.097
◉ Cp / Cv (K1) (SS#1 &SS#2-Inlet/Aftermix)	1.150	1.148	1.132	1.119	1.117	1.115
☒ Compressibility (Z1) (SS#1 &SS#2-Inlet/Aftermix)	.967	.918 / .927	.869 / .864	0.972	.929 / .948	.896/ .928
☒ Inlet Volume, m3/h (Dry)	216753	148,752	102,331	198,944	41,640	11,953

DISCHARGE CONDITIONS	SECT.1	SECT.2	SECT.3
☒ Pressure, (Kg/cm2) A NOTE (4)	3.71	7.33	14.50	4.75	7.58	13.96
☒ Temperature, °C (to be verified by vendor)	9.0	24.7	59.5	80.4	86.6	109.8
☒ Cp / Cv (K2) AVG	1.150	1.145	1.142	1.119	1.117	1.115
☒ Compressibility (Z2) AVG	.950	.903	.844	.964	.941	.911
☒ KW Required (All Losses Included) ± 4 %		39,419			36,465	
☒ Speed, rpm		3063			3059	
☒ Estimated Surge, m3/h (at Speed Above)	SEE PERFORMANCE CURVES					
☒ Polytropic Head, M-kg/kg F.F / IMP.	5441/5613	3382/2737	3476/3960	5674	2959	3941
☒ Polytropic Efficiency, % F.F / IMP.	82.5/85.1	106.1/85.8	71.9/82.0	84.4	90.0	79.6
○ Guarantee Point See note 1, page 13 of 15						
☒ Performance Curve No.	see pages 4, 5, 6, 7 of 12					

PROCESS CONTROL By HCJV

Method : ○ Suction Throttling From ____ (bar) To ____ (bar) ○ Variable Inlet Guide Vanes (3.4.2.4) ◉ Speed Variation From 80 % To 105 % ○ Discharge Blowoff To ____ ◉ Cooled Bypass From Discharge To Suction

Signal : ◉ Source (3.4.2.1) MANUAL CONTROLLER (Electronic Signal/hydraulically Actuated) FOR SPEED CONTROL

Type : ○ Electronic ○ Pneumatic ◉ Other ANTI-SURGE CONTROLLER (Electronic Signal / Pneumatically Actuated) FOR COOLED

Range 4 - 20 mA 0.2 - 1.0 (kg/cm2G)

Antisurge Bypass : ○ Manual ○ Automatic ○ None

REMARKS

(1) AT SIDESSTREAM INLET TO COMPRESSOR

(2) VENDOR SHALL CONFIRM GAS PROPERTIES AT EACH INLET, EACH DISCHARGE AND MIX TEMPERATURES AT SS#1 AND SS#2 INLETS.

(3) VENDOR SHALL CONFIRM COMPRESSOR PERFORMANCE DURING FULL RECYCLE. GAS CONDITIONS UPSTREAM OF EACH COMPRESSOR INLET ANTI-SURGE CONTROL VALVE IS 41.9 °C AND 13.94 KG/CM2 (ABS)

(4) IN RECYCLE CASE SUCTION TEMPERATURE TO BE AT LEAST 10°C ABOVE DEWPOINT. VENDOR TO CONFIRM MAXIMUM TEMPERATURE ACCEPTABLE

(5) DELETED

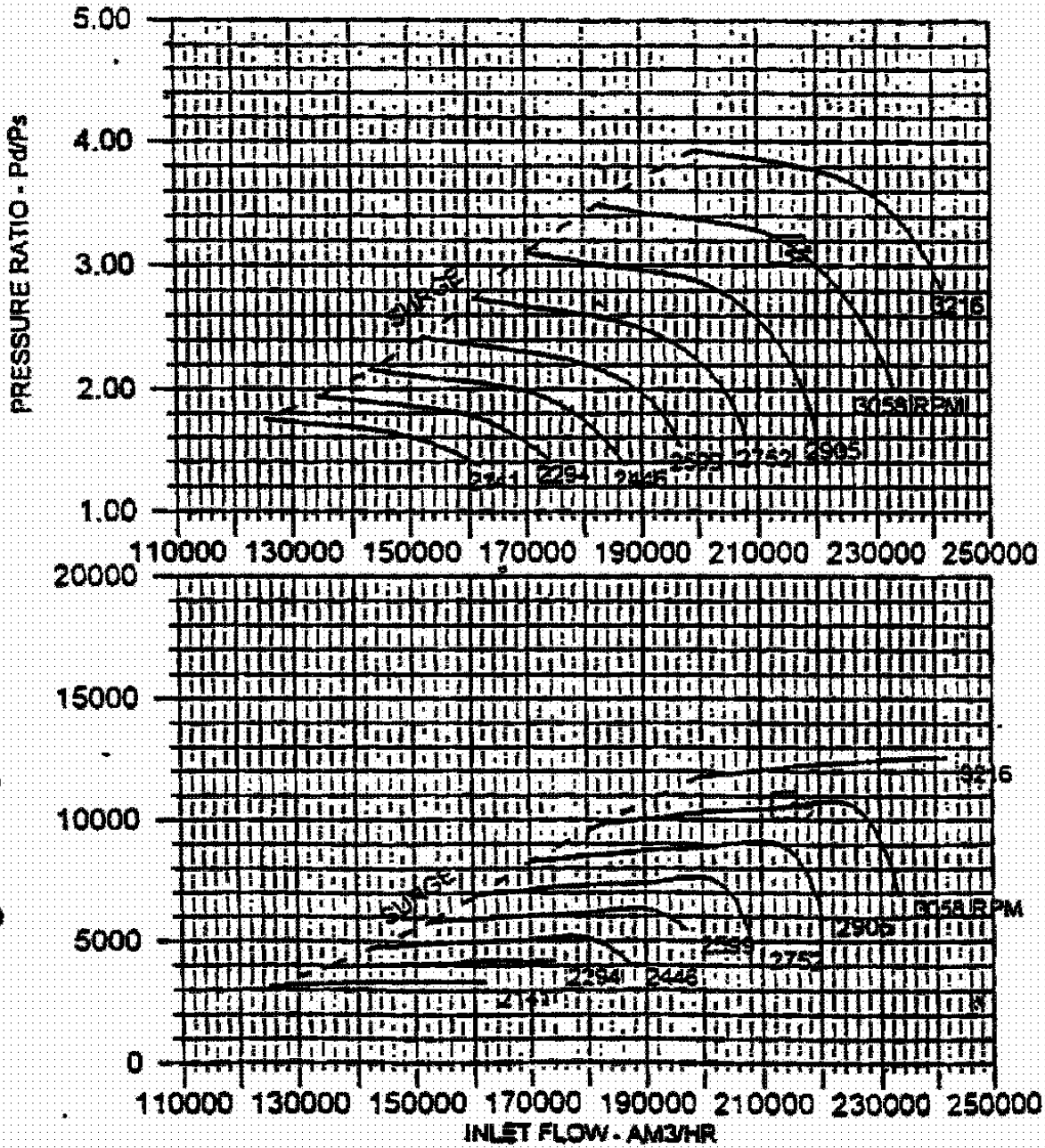
IKPT KELLOGG

CENTRIFUGAL COMPRESSOR
DATA SHEET
SI (METRIC) UNITS

JOB NO. 95038/6413 ITEM NO H4K-1
P O NO PGH-KPO-K-003 DATE 25 May 1998
REQUISITION NO PGH-MR-K-003
DOC. NO. DS-H04-K-001

5

	CASE 1 ○	CASE 2 □	CASE 3 △
PS,KG/CM2A	1.19	1.25	1.17
PD,KG/CM2A	3.75	3.93	3.71
TS,DEG. C	-37.8	-36.7	-38.3
MW	44.1	44.1	44.1
KG/HR	589731.	616538.	580671.
RPM	3058	3026	3063



KW = KWcurve X Ps/1.19

IKPT

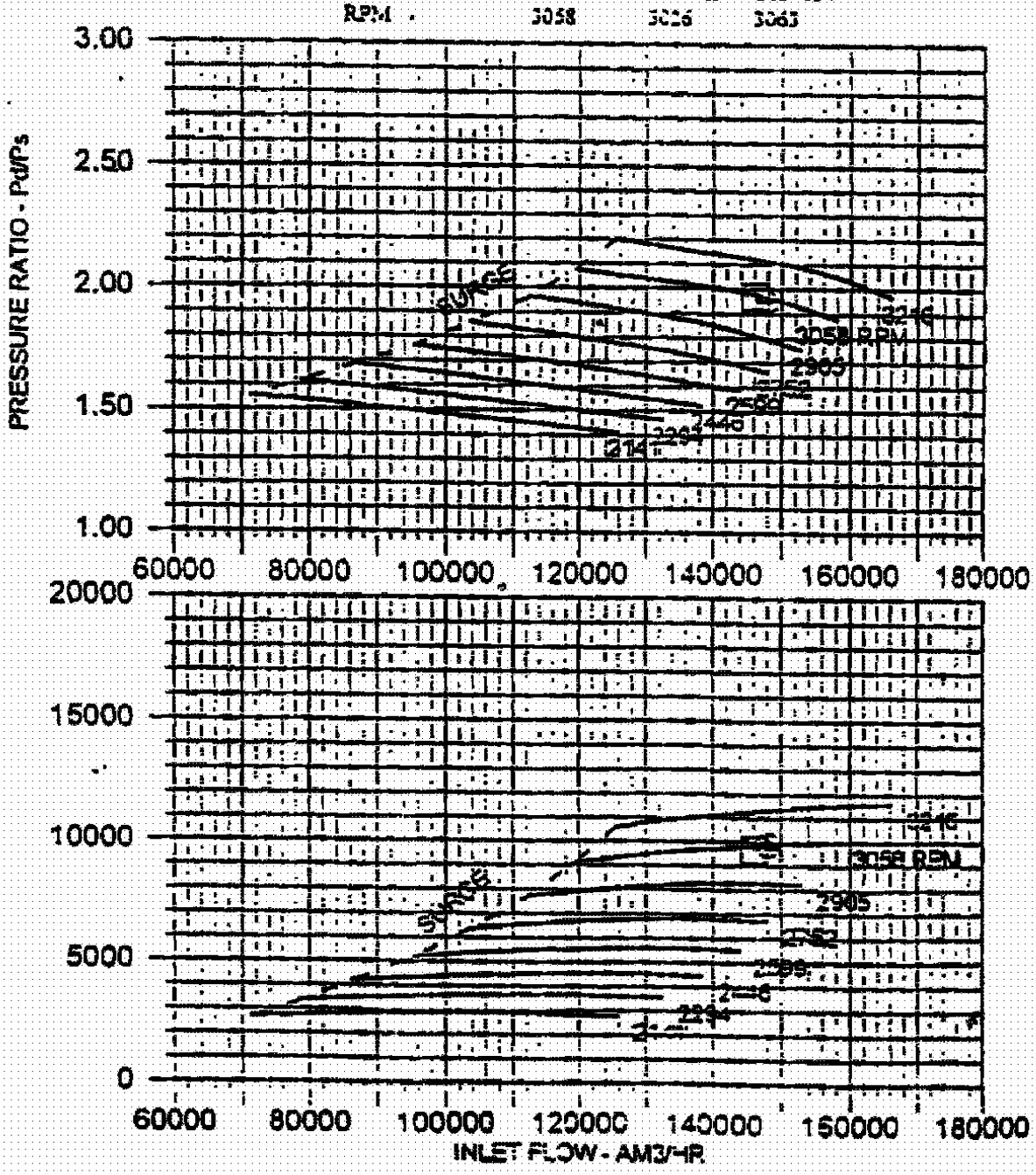
KELLOGG

CENTRIFUGAL COMPRESSOR
DATA SHEET
SI (METRIC) UNITS

JOB NO 95038/64'3 ITEM NO 4K-1
P O NO PGM-KPC-K-003 DATE 29 May 1998
REQUISITION NO PGM-MP-K-003
DOC NO. DS-M04-K-001

5

	CASE 1	CASE 2	CASE 3
PS.KG/CM2A	3.75	3.82	3.71
PD.KG/CM2A	7.35	7.36	7.33
TS.DEG. C	1.2	2.4	1.5
MW	44.1	44.1	44.1
KG/HR	1130478.	1149913	1124834
RPM	3058	3026	3063



KW = KWcurve X Ps/3.75

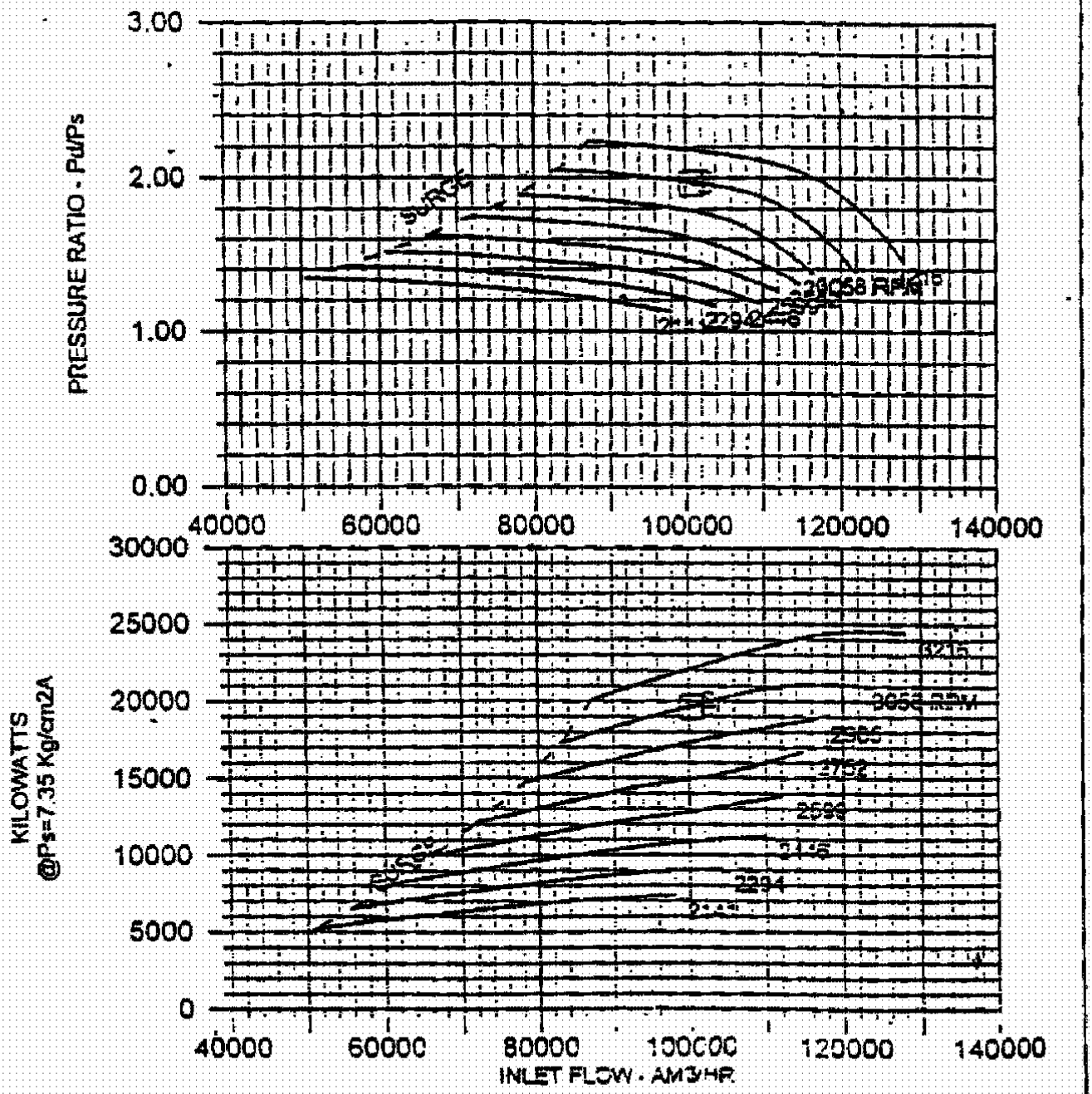
IKPT KELLOGG

CENTRIFUGAL COMPRESSOR
DATA SHEET
SI (METRIC) UNITS

JOB NO 95038/6-[illegible] ITEM NO -MK-1
P O NO PGH-KPC-K-303 DATE 25 May 1998
REQUISITION NO PGH-MR-K-303
DCC NO DS-H04-K-701

5

	CASE 1	CASE 2	CASE 3
PS,KG/CM.A	7.35	7.35	7.35
PD,KG/CM.A	14.5	14.5	14.5
TS,DEG. C	22.7	24.3	22.5
MW	44.1	44.1	44.1
KG/HR	1502505.	1512297	1497513.
RPM	3053	3025	3063



KW = KWcurve X Ps/7.35

IKPT

CENTRIFUGAL COMPRESSOR DATA SHEET SI (METRIC) UNITS

JOB NO. *95038/6413* ITEM NO. *H4K-1*
P.O. NO. *PGH-KPO-K-003* DATE *25 May 1998*
REQUISITION NO. *PGH-MR-K-003*
DOC. NO. *DS-H04-K-001*

OPERATING CONDITIONS (CONTD) (3.1.2 & 3.1.3)

GAS ANALYSIS ◉ MOL % ○	MW	DESIGN & GUARANTEE CASE 1 INLET	CASE 1 SS#1	CASE 1 SS#2	CASE 2 INLET	CASE 2 SS#1	CASE 2 SS#2	CASE 3 INLET	CASE 3 SS#1	CASE 3 SS#2	REMARKS (2.11.1.3 & 2.11.1.8)
Air	28.966										
Oxygen	32.000										
Nitrogen	28.016										
Water Vapor	18.016										
Carbon Monoxide	28.010										
Carbon Dioxide	44.010										
Hydrogen Sulfide	34.076										
Hydrogen	2.016										
Methane	16.042										
Ethylene	28.052										
Ethane	30.068										
Propylene	42.078										
Propane	44.094	100	100	100	100	100	100	100	100	100	
Iso Butane	58.120										
n Butane	58.120										
Iso Pentane	72.146										
n Pentane	72.146										
Hexane Plus											
TOTAL		100	100	100	100	100	100	100	100	100	
Avg Mol Wt		44.097	44.097	44.097	44.097	44.097	44.097	44.097	44.097	44.097	

LOCATION
○ Indoor ○ Heated ◉ Under Roof
◉ Outdoor ○ Unheated ◉ Partial Sides
○ Grade ◉ Mezzanine ○
◉ Electrical Area Class. (2.1.15) Cl I Gr D Div 2
○ Winterization Reqd (2.1.9) • Tropicalization Reqd (3.4.6.6)

SITE DATA
◉ Elevation 14 m ◉ Barometer 1.03 Kg/cm2
◉ Range of Ambient Temp.° C :

	DRY BULB	WET BULB
Site Rated	30°C	28°C
Normal	26°C	24°C
Max.	37°C	
Min.	21°C	

UNUSUAL CONDITIONS ○ Dust ○ Fumes
◉ Other (2.1.9)
SALT LADEN ATMOSPHERE

NOISE SPECIFICATIONS (2.1.10)
◉ Applicable to Machine
See Specification *SPEC A-9*
○ Applicable to Neighborhood
See Specification
Acoustic Housing : ○ YES ◉ NO

APPLICABLE SPECIFICATIONS API Std 617
• Vendor Having Unit Responsibility (2.9.1.7)
◉ Governing Specification
SPEC. K-1 (C.2.7.1)

PAINTING
○ Manufacturer's ◉ Other *SPEC. X-2 & PGG-IS-X-1002 REV. 2*

SHIPMENT (4.4.1)
Domestic • Export • Export Boxing Reqd
• Outdoor Storage More Than 6 Months (4.4.1)
Spare Rotor Assembly Packaged for (4.4.3.10)
Horizontal Storage Vertical Storage

REMARKS

IKPT KELLOGG

CENTRIFUGAL COMPRESSOR DATA SHEET SI (METRIC) UNITS

JOB NO. 95038/6413 ITEM NO. H4K-1
P.O. NO. PGH-KPO-K-003 DATE 25 May 1998
REQUISITION NO. PGH-MR-K-003
DOC. NO. DS-H04-K-001

CONSTRUCTION FEATURES

☒ SPEEDS
Max. Cont 3216 rpm Trip 3538 rpm
Max. Tip Speeds : 269 m/s @ rated Speed
282.5 m/s Max. Cont Speed

☒ LATERAL CRITICAL SPEEDS (DAMPED)

First Critical	1900	rpm	Mode
Second Critical	4300	rpm	Mode
Third Critical	-	rpm	Mode
Fourth Critical	-	rpm	Mode

- Lateral Analysis Required
- Undamped Stiffness Map Required
- Train Torsional Analysis Required (Turbine Driven Train) (2.9.4.5)

☐ TORSIONAL CRITICAL SPEEDS
First Critical rpm
Second Critical rpm
Third Critical rpm
Fourth Critical rpm

☒ VIBRATION
Allowable Test Level (pk-to-pk) 38.1 Microns

ROTATION, VIEWED FROM DRIVEN END

- MATERIALS INSPECTION REQUIREMENTS (4.2.2)
  - Special Charpy Testing (2.11.3) Yes
  - Radiography Required for (See Note-3 Page 13 of 15)
  - Magnetic Partical Required for (See Note-3 Page 13 of 15)
  - Liquid Penetrant Required for

☒ CASING
Model 939 B6-4
Casing Split RADIAL
Material ASTM A516 Gr 70
Thickness, mm 101,6 Corr. Allow., mm 3.18
Max. Working Press. 21.1 kg/cm2G
Max. Design Press. 21.1 kg/cm2G
Test Press., Kg/cm2G: Helium N/A Hydro 31.6
Max. Oper Temp (1) 109.6 °C Min. Oper Temp -38.27 °C
Max. No. of Impellers for casing 4
Max. Casing Capacity, m3/h 292.000
Radiograph Quality • Yes No
Casing Split Sealing O - RING
- System Relief Valve Set Pt (2.2.4) 21.1 kg/cm2G

☒ DIAPHRAGM
Material FABRICATED STEEL

☒ IMPELLERS C.M
No. 4 Diameters 157.48 / 137.16 / 154.9 / 162.56
No Vanes Each Impeller 17 , 19 , 17 , 19

(1) VENDOR SHALL CONFIRM FOR CONTINUOUS RECYCLE CASE.

Type (Open, Enclosed, etc) ENCLOSED
Type Fabrication 5 AXIS MILLED
Material AISI 4330
Min. Yield Strength, Kg/cm2 1 @ 7736 3 @ 6329
Brinnel Hardness : Max. 321 / 302 Min. 269 / 241
Smallest Tip Internal Width, mm 62
Max. Mach No. @ Impeller Eye 0.88
Max. Impeller Head @ Rated Speed, m 3365

☒ SHAFT
Material AISI 4340
Diam @ Impellers, mm 438 Diam @ Coupling, mm 203
Shaft End : Tapered Degree of Angle
Max. Yield Strength, bar 8620
Shaft Hardness (BNH) (Rc) 293 - 331
Stress at Coupling, bar 845

☒ BALANCE PISTON
Material 4140 Area 683224 mm2
Fixation Method INTERFERENCE FIT

☒ SHAFT SLEEVES
At Interstg Close Clearance Points Matl 410 SS
At Shaft Seals Matl MONEL "B" WITH COLMONOY OVERLAY

☒ LABYRINTHS
Interstage :
Type KNIFE EDGE Material ALUMINIUM
Balance Piston :
Type KNIFE EDGE Material ALUMINIUM

SHAFT SEALS
◉ Seal Type (2.8.3) OILRUM
◉ Settling Out Pressure (120PSIG) 844 kg/cm2G
○ Special Operation (2.8.1)
○ Supplemental Device Required for Contact Seals (2.8.3.2)
Type
○ Buffer Gas System Required (2.8.7)
○ Type Buffer Gas
○ Buffer Gas Control System Schematic by Vendor
☐ Type Seals
☒ Inner Oil Leakage Guar. L/day/seal 757 TOTAL
Buffer Gas Required :
☐ Air-Run-In ☐ Other
☐ Buffer Gas Flow (per Seal) :
Norm : kg/min @ bar ΔP
Max. : kg/min @ bar ΔP

☒ BEARING HOUSING CONSTRUCTION
Type (Separate, Integral) SEPARATE Split HORIZONTAL
Material C.S ASTM A 36
Separation Seal, Type Split
Material

IKPT — KELLOGG

CENTRIFUGAL COMPRESSOR DATA SHEET SI (METRIC) UNITS

JOB NO. 95038/6413 ITEM NO. H4K-1
P.O. NO. PGH-KPO-K-003 DATE 25 May 1998
REQUISITION NO. PGH-MR-K-003
DOC. NO. DS-H04-K-001

CONSTRUCTION FEATURES (CONTD)

BEARINGS AND BEARING HOUSINGS

RADIAL

	INLET	EXHAUST
☒ Type	Tilting Pad	Tilting Pad
☒ Manufacturer	D-R	D-R
☒ Length, mm	127	127
☒ Shaft Diam. mm	304.8	304.8
☒ Unit Load (Act. / Allow.), bar	15 / 34.5	15 / 34.5
☒ Base Material	ASTM A36	ASTM A36
☒ Babbt Thickness, mm	1.524	1.524
☒ No. Pads	FIVE	FIVE
☒ Load . Between / On Pad	ON	ON
☒ Pivot : Center / Offset, %	CENTER	CENTER
☒ Bearing Span	4318	4318
☐		

THRUST

	ACTIVE	INACTIVE
☒ Type	Tilting Pad	Tilting Pad
☒ Manufacturer	D-R	D-R
☒ Unit Loading, Max. bar	5.2	5.2
☒ Unit Load, Alt bar	69	69
☒ Area, mm2	84645	84645
☒ No. Pads	14	14
☒ Pivot : Center / Offset, %	55	55
☒ Pad Base Material	ASTM A36	ASTM A36

Lubrication : • Flooded ○ Directed
Thrust Collar : ○ Integral • Replaceable
Material FORGED STEEL

BEARING TEMPERATURE DEVICES

See Attached API 670 Data Sheet
Thermistors
Type - Pos Temp Coeff - Neg Temp Coeff
Temp Switch & Indicator by : - Purch - Mfr
• Thermocouples
• Selector Switch & Indicator by : √ Purch - Mfr
Resistance Temp Detectors
Resistance Matl - - Ohms
Selector Switch & Indicator by : - Purch - Mfr
• Location-Journal Bearing
No. - Each Pad - Every Other Pad 4" per Brg
Other (*) Included 2 Spares
• Location-Thrust Bearing
No. - Each Pad - Every Other Pad 7"" per Brg
Other (**) Included 5 Spares
No. (Inact) - Each Pad - Every Other Pad 2 per Brg
Other
• Monitor Supplied by (3.4.7 4) PURCHASER
Location - Enclosure -
Mfr - Model -
Scale Range - • Alarm ○ Set @ 110 °C
3 • Shutdown Set @ 120 °C Time Delay - sec

VIBRATION DETECTORS

See Attached API 670 Data Sheet
• Type PROXIMITY ○ Model 3300
• Mfr BENTLY NEVADA
• No. at Each Shaft Bearing 4""" Total No. 8
• Oscillator-Detectors Supplied by COMPRESSOR MFR.
• Mfr BENTLY NEVADA ○ Model 3300
Monitor Supplied by (3.4.7.2) PUCHASER
• Location CCR Enclosure
• Mfr BENTLY NEVADA ○ Model 3300
Scale Range • Alarm ○ Set @ 114 mm 3
• Shutdown ○ Set @ 140 mm Time Delay sec

AXIAL POSITION DETECTOR

See Attached API 670 Data Sheet
• Type PROXIMITY ○ Model 3300
• Mfr BENTLY NEVADA • No. Required 3""""
• Oscillator-Demodulator Supplied by COMPR. MFR.
• Mfr BENTLY NEVADA ○ Model 3300
Monitor Supplied by (3.4.7.2) PURCHASER
• Location CCR Enclosure
• Mfr BENTLY NEVADA ○ Model 3300
Scale Range • Alarm ○ Set @ 127 mm
• Shutdown ○ Set @ 254 mm Time Delay sec 3

CASING CONNECTIONS

CONNECTION	• DESIGN APPROVAL REQD (2.11.2.10) 0	☒ SIZE	☒ FACING	• POSITION (2.4.2.1)	○ FLANGED OR STUDDED (2.4.2.1)	◉ MATING FLG & GASKED BY VENDOR (2.4.2.3)	☒ GAS VELOCITY @ MCS m/s Rtd / Max. Allow.
Inlet	Yes	60	Taylor 350# RF	DOWN	FLANGED	Yes	33
Discharge	Yes	36	MSS 300# RF	DOWN	FLANGED	Yes	22.6
1st Side Stream	Yes	30	MSS 300# RF	DOWN	FLANGED	Yes	40.5
2nd Side Stream	Yes	16	ANSI 300# RF	DOWN	FLANGED	Yes	51

(***) INCLUDES 2 SPARES INSTALLED
(****) INCLUDES 1 SPARE INSTALLED

IKPT KELLOGG

CENTRIFUGAL COMPRESSOR DATA SHEET SI (METRIC) UNITS

JOB NO. *95038 / 6413* ITEM NO. *H4K-1*
P.O. NO. *PGH-KPO-K-003* DATE *25 May 1998*
REQUISITION NO. *PGH-MR-K-003*
DOC. NO. *DS-H04-K-001*

☐ OTHER CONNECTONS

SERVICE	NO	SIZE	TYPE
Lube Oil Inlet			
Lube Oil Outlet			
Seal Oil Inlet			
Seal Oil Outlet			
Casing Drains			
Stage Drains			
Vents			
Cooling Water			
Pressure			
Temperature			
Purge for :			
Brg Housing			
Between Brg & Seal			
Buffer Gas			
Solvent Injection			

◉ Individual Stage Drains Required (2.4 3.2)
○ Valved & Blinded
◉ Valved & Blinded & Manifold

☒ ALLOWABLE PIPING FORCES AND MOMENTS

	INLET FORCE N	INLET MOMT N.m	DISCHARGE FORCE N	DISCHARGE MOMT N.m	FORCE N	MOMT N.m
AXIAL						
VERTICAL	*PER*	*API*	*617*	*6TH*	*ED.*	
HORIZ 90°						

	FORCE N	MOMT N.m	FORCE N	MOMT N.m	FORCE N	MOMT N.m
AXIAL						
VERTICAL	*PER*	*API*	*617*	*6TH*	*ED.*	
HORIZ 90°						

○ ACCELEROMETER
○ See Attached API 670 Data Sheet
○ Type ☐ Model *PGH-KPO-K-003*
○ Mfr No. Required
○ Location
○ Oscillator-Demodulators Supplied by
○ Mfr ☐ Model
○ Monitor Supplied by (3.4.7.6)
○ Location Enclosure
○ Mfr ☐ Model
☐ Scale Range ○ Alarm : ☐ Set @ mm/s2
○ Shutdown ☐ Set @ mm/s2 ○ Time Dela sec

ACCESSORIES

COUPLING AND GUARD (3.2)

NOTE : See Rotating Elements - Shaft Ends
○ See API 671 Data Sheet
Coupling Furnished by *Compressor Mfr.*
Manufacturer *Lucas (Bendix)* Type *Contoured Diaphragm Flexible* Model *74T526*
Coupling Guard Furnished by *Compressor Mfr.* No. of Vent Conn : *1* Size, mm *203.2*
Type : ◉ Fully Enclosed ○ Semi-open ○ Oth

Coupling Details :
☐ Max. OD mm
☐ Hub Weight kg
☐ Spacer Length mm
☐ Spacer Weight kg

◉ Vendor Mount Half Coupling
○ Idling Adapter / Solo Plate Reqd (3.2.4)
Lubricating Requirements :
◉ Nonlube ○ Grease ○ Cont Oil Lube ○ Other
Quantity per Hub kg or m3/h
◉ Hydraulic coupling removal / attachment KIT
(Adapter + Hydraulic Pump)

MOUNTING PLATES

◉ Baseplates : Furnished by (3.3.1.1) *Compr. Mfr.*
◉ Compressor Only (3.3.2.1) ○ Driver ○ Gear
○ Other
○ Drip Trim ○ Leveling Pads (3.3.2.2)
○ Column Mounting (3.3.2.3)
○ Subsoleplates Reqd (3.3.3.2)
☒ Stainless Steel Shim Thickness *6.3* mm
◉ Primer for Epoxy Grout Reqd (3.3.1.2.9)
Type *Catalyzed Epoxy Primer*

○ Soleplates : Furnished by
☐ Thickness mm
○ Subsoleplates Reqd (3.3.3.2)
○ Leveling (Chock) Blocks Reqd
☐ Stainless Steel Shim Thickness
○ Driver ○ Gear ○ Compressor
○ Primer for Epoxy Grout Reqd (3.3.1.2.9)
Type

REMARKS

IKPT KELLOGG

CENTRIFUGAL COMPRESSOR DATA SHEET SI (METRIC) UNITS

JOB NO. 95038/6413 ITEM NO. H4K-1
P.O. NO. PGH-KPO-K-003 DATE 25 May 1998
REQUISITION NO. PGH-MR-K-003
DOC. NO. DS-H04-K-001

UTILITIES

○ UTILITY CONDITIONS

STEAM		DRIVERS		HEATING	
Inlet Min.	17.6 kg/cm2	°C	barg		°C
Norm.	17.6 kg/cm2	°C	barg		°C
Max.	19.3 kg/cm2	°C	barg		°C
Exhaust Min.	3.9 kg/cm2	°C	barg		°C
Norm.	3.9 kg/cm2	°C	barg		°C
Max.	3.9 kg/cm2	°C	barg		°C

ELECTRICITY

	DRIVERS	HEATING	CONTROL	SHUTDOWN
Voltage	460	240	120	120
Hertz	50	50	50	50
Phase	3	1	1	1

COOLING WATER

Temp Inlet	34 °C	Max. Return	46		°C
Press.: Norm.	2.1 Kg/cm2G	Design	10.5		Kg/cm2G
Min. Return	1.1 Kg/cm2G	Max. Allow. ΔP	1.0		Kg/cm2G
Water Source		Sea Water			

INSTRUMENT AIR

Normal Press.	7.0	Kg/cm2G		
Max. Press.	9.0 Kg/cm2G	Min. Press.	4.2	Kg/cm2G

SHOP INSPECTION AND TEST (4 1.4)

	REQD	WITNESS	OBSER'D
Shaft Runout	◉	◉	
Shop Inspection	◉	◉	
Cleanlines (4.2.1.5)	○		
QC Program Review	◉		
Hydrostatic	◉	◉	○
Impeller Overspeed	◉	◉	○
Mechanical Run with shop driver	◉	◉	○

☒ Contract Coupling ☐ Idling Adaptor(s)
☒ Contract Probes ☐ Shop Probes

Vary Lube & Seal Oil Pressures and Temperatures (4.3.4.2.5)	◉		
Polar Form Vibration Data (4.3.4.3.3)	◉		
Tape Record Vibration Data (4.3.4.3.6)	◉		
Tape Data to Purchaser (4.3.4.3.7)	◉		
Shaft End Seal Inspection (4.3 4.4.1)	◉	◉	
Gas Leak Test Disch Press. (4.3.5.3)	◉	◉	○
○ Before ◉ After Mech Run			
Performance Test (Gas) (4.3.6.1)	◉	◉	○
Complete Unit Test (4.3.6.2)	○	○	○
Torsional Vibration Meas (4.3.6.2)	○	○	○
Tandem Test (4.3.6.3)	○	○	○
Gear Test (4.3.6.4)	○	○	○
Helium Leak Test (4.3.6.5)	○	○	○
Sound Level Test (4.3.6.8)	◉	◉	○
Full Load/Speed/Press. Test (4.3.6.9)	○	○	○
Hydraulic Coupling Inspection (4.3.6.10)	◉	◉	○
Certified Copies of Test Data	◉	○	○
Disassy. / Reassy. Compressor After Test (3)	◉	◉	○
Preparation for Shipment	◉	◉	○
Residual Elect / Mech Run Out	◉	◉	○
Bearing / Seal Disassembly	◉	◉	
Unbalance Response Verification	◉	◉	

☐ TOTAL UTILITY CONSUMPTION

Cooling Water	m3/h
Steam, Normal	Kg/h
Steam, Max.	Kg/h
Instrument Air,	mm3/h
Horsepower (Driver)	kW
Horsepower (Auxiliaries)	kW
Heaters	kW
Purge (Air or N2)	mm3/h

MISCELLANEOUS

☒ Recommended Straight Run of Pipe Diameters
◉ Before Suction 3 @ Inlet, 4 @ 1st S.S., 7-8 @ 2nd S.S.
◉ Vendor's Review & Comments on Purchaser's Piping & Foundation (3.5.3.2)
○ Compressor to be Suitable for Field Run-in on Air
○ Provision for Liquid Injection (2.1.11)

◉ Vendor's Review & Comments on Purchaser' Control System (3.4.1.1)
○ Extent of Process Piping by Vendor (3.5.3.1)
○ Shop Fitup of Vendor Process Piping (4.4.3.11)
○ Welding Hardness Testing (4.2.1.6)
○ Auxiliary Equipment Motors Explosionproof (3.1.6)
◉ Torsional & Lateral Critical Analysis Report Req'd.

☒ WEIGHTS, kg

Compr 143000 Gear ____ Driver 56300 Base 10900
Rotors : Compr 9100 Driver 10700 Base ____
Compr Upper Case ____
Sour Seal Oil Traps
L. O. Console 48,300 S. O. Console
Overhead Seal Oil Tanks
Max. for Maintenance (Identify) 72700
Total Shipping Weight 155,000 Comp. 56,300 Turbine

☒ SPACE REQUIREMENTS, m

Complete Unit :	L 181	W 41	H 58
L. O. Console :	L 12.1	W 56	H 27
S. O. Console :	L	W	H
Sour Seal Oil Tanks			
Overhead Seal Oil Tanks			

SAFETY CONTROLS (By Contractor)

Control Logic :
Relay - Single - Dual - 2 of 3
PLC - Dual - 2 Out of 3
Vendor to Privide :
Total System - Local Components
PLC :
Interface - Mfr

REMARKS

1. UTILITY CONDITIONS. REFER TO SH-ED-PE-03 (1)
2. OTHER SHOP INSPECTION AND TESTS SHALL BE BE CONDUCTED AS PER PGH-QC-A-002.
3. IF UNSATISFACTORY RUN TEST (2)

IKPT KELLOGG

CENTRIFUGAL COMPRESSOR DATA SHEET SI (METRIC) UNITS

JOB NO. 95038/6413 ITEM NO. H4K-1
P.O. NO. PGH-KPO-K-003 DATE 25 May 1998
REQUISITION NO. PGH-MR-K-003
DOC. NO. DS-H04-K-001

SPECIAL NOTES ;

(1) COMPRESSOR SUCTION DATA FOR MEDIUM AND HIGH PRESSURE STAGES IS OBTAINED BY COMBINING THE DISCHARGE FROM THE PREVIOUS STAGE WITH AN EXTERNAL SIDE STREAM.

(2) THE ACCEPTANCE CRITERIA FOR THE SHAFT HORSE POWER INCLUDING ALL LOSSES AT THE GUARANTEE POINT (CASE 1 - NORMAL) SHALL NOT EXCEED 104% OF THE SPECIFIED VALUE DURING THE PTC-10 SHOP PERFORMANCE TEST.

(3) MATERIAL TEST, SHOP INSPECTION AND TEST REQUIREMENTS OTHER THAN THOSE SPECIFIED IN THIS DATA SHEET SHALL BE REFERRED TO PGH-QC-A-002 " SHOP INSPECTION AND TEST PLAN".

(4) COMPRESSOR AND TURBINE VENDOR SHALL DEMONSTRATE INTERCHANGEABILITY OF THE COMPLETE BUNDLE (ROTOR, DIAPHRAGMS, ETC) WITH EXISTING 4K-1 CASES (IE. IN THE EXISTING TRAIN G) . VENDOR TO PROVIDE A DIMENSIONAL RECORD TO HCJV (PURCHASER) FOR REVIEW AND APPROVAL.

(5) SEAL AND BEARING PARTS MUST BE INTERCHANGEABLE WITH THE EXISTING 4K-1's TRAIN-G.

(6) SHAFT COUPLING SHALL BE RATED ACCORDING TO THE PERFORMANCE REQUIREMENTS FOR OPERATING CONDITIONS SPECIFIED ON PAGES 2 AND 3.
VENDOR TO PROVIDE A SHAFT WITH UPGRADED HEAT TREATMENT TO INCREASE SERVICE FACTOR FROM 1.13 TO BE 1.4.
FINAL SHAFT SERVICE FACTOR AT MCS AT MAXIMUM DRIVER BHP MUST EQUAL OR EXCEED EXISTING (TRAIN-G) SHAFT SERVICE FACTOR.
VENDOR TO PROVIDE CALCULATIONS OF COUPLING HUB TO SHAFT FIT AREA TO CONFIRM SERVICE FACTOR AT MCS AND MAXIMUM DRIVER (RATED) POWER EQUALS OR EXCEED TRAIN-G COUPLING/ SHAFT JOINT SERVICE FACTOR (1.4 MINIMUM)

(7) OTHERS :

a. TORSIONAL AND LATERAL CRITICAL ANALYSIS REPORT REQUIRED WITH COMPLETE ROTOR DYNAMICS AND BEARING STABILITY REPORT PER SPECIFICATION

b. VENDOR TO TERMINATE ALL WIRING AT TERMINAL BOXES AT EDGE OF SKID

c. OPTICAL ALIGMENT PADS (BENCH MARKS) REQUIRED ON COMPRESSOR / DRIVER SKID

d. EXCITATION RESONANT TESTING IS REQUIRED ON EACH H4K-1 COMPRESSOR IMPELLER.

e. IMPELLERS TO BE 5 AXIS MILLED, DIAPHRAGMS TO BE FABRICATED (SEE NOTE 7.6, PAGE 18 OF 18 ON REQUISITION PGH-MR-K-003)
FIRST STAGE IMPELLER TO HAVE UPGRADED HEAT TREATMENT TO INCREASE MINIMUM MATERIAL YIELD STRENGTH

f. ACCEPTABLE PRE-RUN DATA IS REQUIRED BEFORE HCJV TRAVELS TO WITNESS THE MECHANICAL RUN TEST.

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COUPLING (API 671-2ND)
DATA SHEET
SI UNITS

JOB NO 95036 / 6413 ITEM NO. H4K-1
PURCHASE ORDER NO. PGN-PO-K-003
REQUISITION NO. PGN-MR-K-003
REVISION NO. 6 DATE 25 May 1998
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APPLICABLE TO: ○ PROPOSAL ● PURCHASE ○ AS BUILT DATE 25 May 1998 REVISION 6

● FOR BONTANG LNG EXPANSION PROJECT, TRAIN - H NO.REQUIRED 1 (ONE)
SITE BONTANG, EAST KALIMANTAN, INDONESIA SERVICE PROPANE COMPRESSOR

● DRIVER TYPE STEAM TURBINE MANUFACTURER GENERAL ELECTRIC MODEL DRV 1334
SERIAL NO. TAG. NO.
DRIVEN UNIT COMPRESSOR MANUFACTURER DRESSER RAND MODEL 939 B6-4
SERIAL NO TAG. NO.

■ COUPLING: TYPE CONTOURED DIAPHRAGM FLEXIBLE
MANUFACTURER LUCAS (BENDIX) MODEL 74T526 SIZE 526
ASSEMBLY DRAWING NO 74 F 526 - 6185 5

NOTE INFORMATION TO BE COMPLETED ○ BY PURCHASER; ☐ BY COUPLING MANUFACTURER.

OPERATING CONDITIONS

● POWER TRANSMITTED (kW): NORMAL 39480 5 MAXIMUM CONTINUOUS
● SPEED (RPM) MINIMUM ALLOWABLE 2430 NORMAL 3056 MAXIMUM CONTINUOUS 3216 TRIP 3538
○ TORQUE (N-m). MAXIMUM CONTINUOUS MAXIMUM TRANSIENT
○ FREQUENCY OF TRANSIENTS (EVENTS/TIME) (2.1.1)
○ ADDITIONAL ANF EXCITING FREQUENCY (2.3.4.C)
● MINIMUM REQUIRED EXPERIENCE FACTOR, SF (2.1.1) 1.4 ■ ACTUAL SF 1.83 5
● AMBIENT TEMPERATURE (°C) MAXIMUM 37 MINIMUM 21
ENVIRONMENT (2.6.5) (2.1.6) ○ DUST ○ HYDROGEN SULFIDE ● SALT LADEN ATMOSPHERE

LUBRICATION

○ CONTINUOUS (2.2.7 & 8) ○ FILTRATION MICRONS ○ VISCOSITY (cP) @ °C
○ BATCH (2.2.7) ● NONLUBRICATED ○ PRESSURE (kPa) (BAR) NORMAL MAXIMUM
○ GREASE-PACKED ○ TEMP. (°C) NORMAL MAXIMUM
○ OILFILLED ☐ FLOW (m³/h) NORMAL MAXIMUM

COUPLING DATA

● SHAFT SEPARATION AT OPERATING TEMP. (mm B.S.E.) (1) THERMAL GROWTH (mm) FROM °C AMBIENT
○ MOTOR ROTOR FLOAT (mm) ○ LIMITED END FLOAT (mm) (2.2.3) ○ ELECTRICALLY INSULATED (2.1.12)
○ MARINE TYPE REQUIRED (2.2.2) ○ FLEX-HUB TYPE REQUIRED (2.2.2)
○ REQUIRED MISALIGNMENT CAPABILITY (2.1.2)
STEADY STATE ANGULAR (DEG.) PARALLEL OFFSET (mm) AXIAL(mm)
TRANSIENT ANGULAR (DEG.) PARALLEL OFFSET (mm) AXIAL(mm)
○ DYNAMIC BALANCE (2.6.2.1) ● COMPONENT BALANCE & ASSEMBLY CHECK BALANCE (2.6.2.1) (2.6.2.3)
○ ASSEMBLY BALANCE (2.6.2.4) ○ REPEATABILITY TEST AFTER ASSEMBLY BALANCE (2.6.2.5)
○ RESIDUAL UNBALANCE CHECK OF COMPONENTS (2.6.2.2)
■ DESIGN RATING (kW/100 RPM) (2.1.1) 2497 ■ MAXIMUM CONTINUOUS TORQUE (N-m) (5.2.5.8c) 238450 5
■ BOLTING TORQUE (N-m) (2.1.11.5) 53.7/305.1 ☐ LUBRICATED ☐ DRY
■ MAXIMUM ALLOWABLE MISALIGNMENT (5.2.5.8a)
STEADY STATE: ANGULAR (DEG.) 0.200 PARALLEL OFFSET (mm) 4.03 AXIAL(mm) 5.56
TRANSIENT: ANGULAR (DEG.) 0.200 PARALLEL OFFSET (mm) 4.03 AXIAL(mm) 5.56
■ MAXIMUM ALLOWABLE RESIDUAL UNBALANCE (g-mm) DRIVER END 3500 DRIVEN UNIT END 3286
☐ MAXIMUM ACTUAL RESIDUAL UNBALANCE (g-mm) DRIVER END DRIVEN UNIT END
■ TORSIONAL STIFFNESS (kg-mm/RAD) (5.2.5.2b) 1265.76×10^4 WR²(kg-m²) (5.2.6.2b) 5
■ FLEXIBLE-ELEMENT COUPLING INITIAL DEFLECTION (mm) 3.81 ☐ PRESTRETCH ☐ COMPRESSION
■ FLEXIBLE-ELEMENT COUPLING CALCULATED AXIAL NATURAL FREQUENCY 3558 - 4965 CPM
○ TEST OF ANF (4.4.1) ☐ ACTUAL ANF

(1) DUPLICATE TO TRAIN - G

NOTE VENDOR TO COMPLETE THE DATA IN THE BLANK CELLS

IKPT KELLOGG

COUPLING (API 671-2ND)
DATA SHEET
SI UNITS

JOB NO. 95035/6413 ITEM NO. HHK-1
REVISION NO. 6 DATE 29 May 1998
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COUPLING DATA (CONTINUED)

● REQUIRED CURVES (5.2.4.1) (5.2.4.2)
- ● AXIAL FORCE VS AXIAL DEFLECTION
- ● AXIAL FORCE VS TORQUE
- ● BENDING MOMENT VS AXIAL DEFLECTION
- ● BENDING MOMENT VS ANGULAR MISALIGNMENT
- ○ WINDAGE LOSS VS COUPLING SPEED

MATERIALS (2.6)

	DRIVE END MATERIALS	DRIVEN END MATERIALS
5 HUB/FLANGE	AISI 4340	
SPACER	AISI 4130 / 4340	
SLEEVE	N.A	SAME
FLEXIBLE-ELEMENT	AMS 6414	AS
FLEXIBLE-ELEMENT GUARD	AISI 4140	DRIVE
BOLTS	Alloy Steel	END
5 NUTS	Stainless Steel	

● FLEXIBLE-ELEMENT PROTECTIVE COATING (2.6.4) ● VENDOR STANDARD ○
☐ MATERIAL AND WELD HARDNESS TESTS REQUIRED (4.3.1.3)
☐ INTERNAL TEETH HARDNESS (ROCKWELL C) (2.2.6) DRIVE END ACTUAL DRIVEN END ACTUAL
☐ EXTERNAL TEETH HARDNESS (ROCKWELL C) (2.2.6) DRIVE END ACTUAL DRIVEN END ACTUAL

COUPLING HUB MACHINING

	DRIVE END	DRIVEN END
● TYPE (INTEGRAL, CYLINDRICAL, TAPER) (2.1.4) (2.1.8.1)	INTEGRAL	
● KEYED OR HYDRAULICALLY FITTED (2.1.8.1)		HYDRAULIC FIT
○ TAPER (1 DEG I.A., 1/2 IN. PER FT., 3/4 IN. PER FT.) (2.1.9.1.1) (2.1.9.1.2) (2.1.9.1.3)		
☐ KEYWAY DIMENSIONS AND NUMBER (2.1.9.2)		
☐ NOMINAL BORE DIAMETER		
☐ INTERFERENCE FIT (mm) MAX./MIN. (2.1.8.1)	/	/
○ PULLER HOLES		
○ BALANCING HOLES (2.6.3)		

COUPLING GUARD (APPENDIX E)

- ○ COORDINATOR (E.2.1)
- ● FLANGED CYLINDRICAL (E.4.1) ○ BASE MOUNT (E.3.2)
- ○ AIR TIGHT (E.4.7.1) ● OIL TIGHT (E.2.7)
- ● SPARK RESISTANT (E.2.8)
- ○ TRANSPARENT WINDOW FOR EACH OIL SPRAY POINT (E.4.6.4)
- ● VENT CONNECTION (E.4.3) (1)
  - ○ 1 INCH NPT WITH FILTER-BREATHER
  - ○ 1 INCH FLANGED, RATING & FACING
- ○ PURGE GAS ○ DRY AIR PURGE (E.4.4)
- ☐ PURGE CONNECTION SIZE & TYPE
- ☐ PURGE m³/h REQUIRED
- ○ OIL MIST COOLING ○ GAS COOLING
- ○ CONTRACT GUARD TO BE USED DURING SHOP TEST (E.4.5)
- ○ ADDITIONAL GUARD DETAILS (2.3.5)

ACCESSORIES

- ○ PRIME EQUIPMENT SUPPLIER TO FURNISH ONE SET OF PLUG AND RING GAGES (APPENDIX D)
- ○ DRILL TEMPLATE FOR INTEGRAL FLANGED HUBS BY (2.1.5)
  - ○ COUPLING MANUFACTURER ○ PURCHASER
- ○ TWO-PIECE STOP RINGS BY COUPLING MANUFACTURER (3.3)
- ● HYDRAULIC INSTALLATION/REMOVAL TOOLING (TO INCLUDE HAND PUMP(S), PRESSURE GAUGE(S), FITTINGS AND HOSE(S)) BY (3.1)
  - ○ PURCHASER ○ COUPLING MANUFACTURER
- ○ PULLER BY COUPLING MANUFACTURER (3.4)
- ● MOMENT SIMULATOR (2.1.7.1) FOR TURBINE DRIVER

APPLICABLE SPECIFICATIONS

- ● API-671, SPECIAL PURPOSE COUPLINGS
- ○

(1) DUPLICATE TO TRAIN - G

PREPARATION FOR SHIPMENT

- ● OUTDOOR STORAGE FOR MORE THAN 6 MONTHS (4.5.2)
- ● EXPECTED STORAGE TIME
- SHIPPING: ○ DOMESTIC ● EXPORT (4.5.3)
- STORAGE: ○ INDOOR ● OUTDOOR (4.5.2)
- ○ SEE COUPLING PURCHASE ORDER FOR PRESERVATION, BOXING, AND SHIPPING INSTRUCTIONS

○ COORDINATION MEETING ATTENDANCE REQ'D (5.2.2)